

Coupon Influence Maximization under Non-Replicable Diffusion

Anonymous Author(s)

ABSTRACT

Classical influence maximization assumes copyable information that can branch from one activated user to multiple neighbors. Digital coupons instead move as non-replicable objects: a recipient redeems, discards, or forwards a coupon to one neighbor, and only redemption creates an adopter. We formalize this setting as coupon influence maximization (CIM), which allocates a limited number of independently diffusing coupons to maximize the expected number of distinct adopters. We prove that CIM is NP-hard and show that adoption spread is monotone and has diminishing returns over integer coupon allocations. Exact greedy consequently achieves a $(1 - 1/e)$ approximation, but its marginal gains are expensive to evaluate. We therefore develop k -joint reverse reachable samples that preserve coupon identities and separate root adoption from reverse reachability. The resulting coverage events yield unbiased estimators of spread and marginal gain. Based on these estimators, CIM-RIS returns a $(1 - 1/e - \epsilon)$ -approximate allocation with high probability and runs in expected near-linear time in the graph size when the budget, accuracy, and spread lower bound are fixed. Experiments on five graph instances derived from real networks show competitive solution quality and practical scalability. Across two graphs and three scenarios with high-precision Monte Carlo greedy references, the observed gap is 0.3–11.1%; on Netscience, it is 1.1–4.4%. CIM-RIS outperforms both degree and PageRank baselines in all 72 graph-scenario-budget configurations, while conditioned root sampling improves the variance-time product by 1.3× and 1.5× at $k = 10$ on Netscience and EmailEnron.

CCS CONCEPTS

• Information systems → Social networks; • Theory of computation → Approximation algorithms analysis.

KEYWORDS

influence maximization, coupon diffusion, random walk, reverse influence sampling, submodular optimization

ACM Reference Format:

Anonymous Author(s). 2026. Coupon Influence Maximization under Non-Replicable Diffusion. In *Anonymous Submission*. ACM, New York, NY, USA, 16 pages.

1 INTRODUCTION

Word-of-mouth and viral marketing are important strategies for promoting product adoption in online social networks. Influence maximization (IM) asks how to select a limited number of seed users whose diffusion activates as many users as possible [8]. Over the past two decades, IM has been extensively studied under the Independent Cascade (IC) and Linear Threshold (LT) models, where an activated user may influence multiple neighbors and thereby create

a branching cascade [1, 4, 21]. Many promotional objects, however, are non-replicable and cannot be described by this broadcast assumption. A digital coupon, for example, can be redeemed, discarded, or forwarded to one friend, but it cannot be simultaneously copied to many neighbors. Its diffusion is therefore a random-walk-like path rather than a cascade.

Consider a company that distributes k digital coupons through a social network. When a user receives a coupon, the user may redeem it, discard it, or forward it to one out-neighbor. Redemption and discard both terminate the coupon, whereas forwarding moves the same coupon to the next user without creating another copy. A user is counted as an adopter only after redeeming at least one coupon; users who merely forward coupons do not contribute to the campaign objective. Different coupons diffuse independently, and several coupons may be redeemed by the same user, but that user is counted only once. The company may also assign multiple coupons to the same initial user, since independent coupons starting from that user can follow different paths.

These dynamics raise a seed-allocation problem that is distinct from classical IM. Placing many coupons at individually strong seeds may cause their trajectories to overlap at the same adopters, while placing coupons at poor seeds may result in excessive discards. We therefore define *coupon influence maximization* (CIM): given a budget of k coupons, allocate them to seed users so as to maximize the expected number of distinct adopters. We also consider coupon usage rate, the expected fraction of issued coupons that are eventually redeemed, as a complementary campaign measure.

We formalize the coupon diffusion process and establish the computational and structural properties of CIM. We prove that CIM is NP-hard and show that its adoption-spread objective has diminishing returns over coupon allocations. This property suggests a greedy allocation strategy, but direct marginal evaluation by forward Monte Carlo simulation is too expensive for large networks. Inspired by reverse influence sampling (RIS), we construct coupon-aware reverse reachable (RR) samples and convert marginal evaluation into a coverage computation. The resulting CIM-RIS algorithm achieves a $(1 - 1/e - \epsilon)$ approximation with high probability.

CIM differs from classical IM at both the model and algorithmic levels. In IC and LT, activation and propagation are coupled: an activated user may continue the cascade through multiple neighbors. In CIM, propagation follows one forwarding path, and an intermediate forwarder remains a non-adopter. Algorithmically, a classical RR set represents one broadcast realization, whereas a CIM sample must preserve the independent realization and identity of every coupon. Consequently, neither classical RR-set generation nor its induced maximum-coverage procedure can be applied directly.

In summary, our contributions are threefold. First, we introduce the coupon influence model and formally define CIM, capturing non-replicable single-path diffusion, independent coupons, and overlap among adopters. Second, we prove the NP-hardness of CIM and establish the diminishing-return property of its adoption spread

over integer coupon allocations, which supports a $(1 - 1/e)$ greedy approximation when marginal gains are exact. Third, we develop coupon-aware RR sampling and the scalable CIM-RIS algorithm, which estimates these marginal gains and retains a $(1 - 1/e - \epsilon)$ approximation guarantee with high probability. Experiments on five graph instances derived from real networks evaluate its solution quality, scalability, sensitivity to the sampling budget, capacity-aware placement, and the variance reduction obtained by conditioned root sampling. In particular, CIM-RIS is competitive with Monte Carlo greedy, consistently improves over topology-only seed rankings, and selects 200 seeds on the 154,907-node Douban instance in 31.2 seconds using the validated sample budget.

2 RELATED WORK

Influence maximization. Early studies established viral marketing as an optimization problem over social networks [4, 18]. Kempe et al. [8] formalized IM under the IC and LT models, proved NP-hardness, and used monotonicity and submodularity to obtain a $(1 - 1/e)$ greedy approximation. Subsequent work accelerated marginal evaluation through lazy greedy methods and model-specific optimizations [3, 6, 10]. These models describe copyable influence: an activated node can attempt to activate several neighbors. CIM instead allocates non-replicable coupons, each of which follows one forwarding path and creates an adopter only when a user redeems it.

Reverse influence sampling. Borgs et al. [1] introduced the RR-set approach that converts influence estimation into random-set coverage. TIM/TIM+ [23], IMM [21], Stop-and-Stare [7, 16], and OPIM-C [22] progressively improved the theoretical and practical efficiency of this framework. Their fundamental coverage event is that a seed set intersects one RR set generated from a broadcast realization. Our reverse construction differs in two ways: adoption at the root is a separate gate, and one sample contains independent RR components for the individual coupons. This indexed construction is necessary to estimate the distinct adopters created by a coupon allocation.

Diffusion variants. IM has also been studied under competitive diffusion, misinformation control, location constraints, adaptive seeding, and multi-round campaigns [2, 11, 14, 19, 20]. Most directly related to our application, Liao et al. [12] use coupon distribution as the motivation for allocating promotional seeds across rounds to improve a product’s popularity ratio. Their PA-IC model nevertheless propagates copyable promotional information through an IC cascade; it does not model each coupon as the single object being forwarded and consumed. Random-walk processes provide a classical foundation for single-path movement on graphs [13], but standard random-walk objectives do not capture the combination of terminal adoption, non-replicability, and overlap among independently issued coupons. CIM brings these features into one optimization model and develops a coupon-aware reverse sampler for it.

3 MODEL AND PROBLEM DEFINITION

This section introduces the coupon influence model, which combines non-replicable coupons with single-path, random-walk-like

propagation in a social network. We then formulate Coupon Influence Maximization (CIM), which allocates k coupons to initial recipients to maximize the expected number of distinct adopters.

3.1 Coupon Influence Model

Classical IM is commonly studied under the Independent Cascade (IC) and Linear Threshold (LT) models [8]. At time step 0, all seed nodes are activated and the remaining nodes are inactive; once activated, a node remains activated. Under IC, a node activated at time step i has one chance to activate each inactive out-neighbor at time step $i + 1$, independently according to the propagation probability on the connecting edge. Under LT, each node independently selects a threshold from $[0, 1]$ and becomes activated once the total weight from its activated in-neighbors reaches that threshold. The propagation terminates when no more nodes can be activated. Scalable IM algorithms typically solve this problem using reverse influence sampling (RIS) [1, 21]. A random reverse reachable (RR) set is generated by first sampling a root node uniformly from V and then tracing backward all nodes that can activate the root in a random diffusion realization. For any seed set S , its expected influence equals $n \Pr[S \cap R \neq \emptyset]$; hence, IM can be transformed into a maximum coverage problem over sampled RR sets. Given θ independent RR sets, the fraction covered by S gives both the maximum-likelihood estimate and an unbiased estimate of its normalized influence. The required value of θ is determined adaptively using an estimated lower bound on the optimal influence together with concentration bounds, after which a greedy algorithm selects the k nodes with the largest marginal RR-set coverage. CIM follows the same high-level optimization route, but its single-path, non-replicable coupon diffusion requires a different RR-set generation process and a different induced coverage problem.

We model the social network as a directed graph $G = (V, E)$, where V is the set of $n = |V|$ users and E is the set of $m = |E|$ directed forwarding relationships. Let $V_s \subseteq V$ be the set of eligible seed users; we use $V_s = V$ when every user is eligible. Each $v \in V_s$ has a placement capacity $c_v \in \{1, \dots, k\}$. Setting $c_v = 1$ requires distinct seeds, while $c_v = k$ permits all coupons to be assigned to the same user. We assume $\sum_{v \in V_s} c_v \geq k$. Each node $u \in V$ has an adoption probability p_u^a , a discard probability p_u^d , and a transfer probability p_u^t , where $p_u^a + p_u^d + p_u^t = 1$. For every edge $(u, v) \in E$, $p(u, v)$ denotes the unconditional probability that u transfers its coupon to v ; accordingly, $\sum_{v: (u, v) \in E} p(u, v) = p_u^t$. For a node with no out-neighbor, we require $p_u^t = 0$.

Consider one coupon seeded at $s \in V_s$. At time step 0, node s holds the coupon. If node u holds it at time step τ , then u redeems the coupon and becomes an adopter with probability p_u^a , or discards it with probability p_u^d ; either action terminates this coupon’s diffusion. Otherwise, u transfers the coupon to out-neighbor v with probability $p(u, v)$, and v holds it at time step $\tau + 1$. Thus, a coupon generates one forwarding path rather than a branching cascade.

Propagation and adoption are separate in this process. A forwarding user may help the coupon reach a later adopter without becoming an adopter, while a user contributes to the objective only by redeeming the coupon. This separation is absent from classical IC and LT diffusion, where an activated node can continue propagating influence.

We use a realization-space view to analyze the random process. For every coupon $j \in [k]$, an independent realization W_j fixes the action taken by every node for that coupon. Specifically, for each node $u \in V$ and coupon j , we draw an independent random number $r_{u,j} \sim \text{Uniform}[0, 1]$. We use this number to choose the action of u by partitioning the unit interval into one adoption segment of length p_u^a , one discard segment of length p_u^d , and one transfer segment for each out-neighbor v , whose length is $p(u, v)$. Let $\Phi_j(u)$ denote the resulting action of u in realization W_j . Then $\Phi_j(u) = \text{adopt}$ means that u redeems coupon j , $\Phi_j(u) = \text{discard}$ means that u discards it, and $\Phi_j(u) = v$ means that u transfers it to out-neighbor v . The collection $W_j = \{\Phi_j(u)\}_{u \in V}$ is the realization of coupon j , and $\mathcal{W} = (W_1, \dots, W_k)$ fully determines the outcome of all coupons.

Under this construction, coupon j seeded at s_j is redeemed by node u if and only if it reaches u under realization W_j and u takes the adoption action in that realization. Let $A_{u,j}$ denote the event $r_{u,j} \leq p_u^a$. We write

$\text{Adopt}_{u,j}(s_j) = A_{u,j} \wedge \{\text{coupon } j \text{ starting from } s_j \text{ reaches } u \text{ in } W_j\}$.

Here $A_{u,j}$ depends on the action of the target node, whereas the reachability event depends on the actions of preceding nodes on the forwarding path. The construction therefore separates reaching a user from adoption by that user.

For a campaign with k coupons, let $S = (s_1, \dots, s_k)$ be an ordered seed tuple, where coupon j starts from s_j . The coupons diffuse independently and do not block or modify one another. Each coupon follows its own path until it is redeemed, discarded, or trapped in a transfer cycle. Different coupons may reach the same user, who may redeem more than one of them; that user contributes one adopter to the objective regardless of the number of coupons redeemed. Consequently, the coupons are independent at the diffusion level but coupled through distinct-user coverage in the campaign objective.

For a seed tuple $S = (s_1, \dots, s_k)$, node u becomes an adopter if

$$\text{Adopt}_u(S) = \bigvee_{j=1}^k \text{Adopt}_{u,j}(s_j).$$

Comparison and boundary cases. The coupon model differs from related diffusion and simulation settings in three respects.

- *Classical influence maximization.* IC and LT diffusion models describe copyable influence: once a user is activated, the influence attempt may branch to multiple neighbors. In CIM, each coupon is non-replicable and moves to at most one out-neighbor at each step, so the diffusion trace of one coupon is a path rather than a cascade.
- *Forward Monte Carlo evaluation.* Forward Monte Carlo simulation can estimate the adoption spread of a fixed seed tuple by sampling coupon trajectories. Optimization, however, requires repeated marginal evaluation over many allocations, making direct simulation expensive.
- *Repeated visits and coupon overlap.* The realization-space view fixes the behavior of each node for each coupon. Therefore, if the same coupon revisits a node through transfer actions, it follows a deterministic transfer cycle and contributes no adoption in that realization. Different coupons use independent realizations and may be redeemed by the

same user, but the adoption spread counts that user only once.

Challenge. The central algorithmic challenge is to estimate marginal gains without repeated forward simulations while preserving independent coupon realizations and distinct-user coverage.

3.2 Problem Formulation

For an ordered seed tuple $S = (s_1, \dots, s_k) \in V_s^k$, define its allocation vector $\mathbf{x}(S) \in \mathbb{Z}_{\geq 0}^{V_s}$ by $x_v(S) = |\{j : s_j = v\}|$. Thus $\sum_{v \in V_s} x_v(S) = k$; repeated seed nodes are allowed whenever their capacities exceed one. Feasible allocations also satisfy $x_v \leq c_v$. Because coupons diffuse independently according to the same model, the expected outcome depends on S only through $\mathbf{x}(S)$.

We define the adoption spread as the expected number of distinct users who redeem at least one coupon:

$$\sigma(S) = \sum_{u \in V} \Pr[\text{Adopt}_u(S)] = \sum_{u \in V} \Pr\left[\bigvee_{j=1}^k \text{Adopt}_{u,j}(s_j)\right].$$

For nodes $v, u \in V$, let q_{vu} be the probability that one coupon seeded at v is eventually redeemed by u . Independence among coupons gives the equivalent allocation form

$$\sigma(\mathbf{x}) = \sum_{u \in V} \left(1 - \prod_{v \in V_s} (1 - q_{vu})^{x_v}\right).$$

The product is the probability that none of the allocated coupons is redeemed by u . This form makes explicit that each adopter is counted once.

We also define the expected total number of redeemed coupons as

$$\rho(S) = \sum_{j=1}^k \Pr\left[\bigvee_{u \in V} \text{Adopt}_{u,j}(s_j)\right],$$

where the inner event means that coupon j is eventually redeemed by some user. Since each coupon can be redeemed at most once, $0 \leq \rho(S) \leq k$. Equivalently,

$$\rho(\mathbf{x}) = \sum_{v \in V_s} x_v \sum_{u \in V} q_{vu}.$$

The corresponding coupon usage rate is

$$\eta(S) = \frac{\rho(S)}{k}, \quad \eta(\mathbf{x}) = \frac{\rho(\mathbf{x})}{k}.$$

The two quantities capture different goals: σ emphasizes distinct adopters, whereas η emphasizes coupon utilization.

Definition 3.1 (CIM: Adoption-Spread Maximization). Given (a) a directed social network $G = (V, E)$, (b) an eligible seed set $V_s \subseteq V$ with capacities $\mathbf{c}$, (c) node behavior parameters $\{(p_u^a, p_u^d, p_u^t)\}_{u \in V}$, (d) edge forwarding parameters $\{p(u, v)\}_{(u,v) \in E}$, and (e) a coupon budget k , the task of Coupon Influence Maximization (CIM) is to find a coupon allocation $\mathbf{x}^*$ of total size k that maximizes adoption spread:

$$\mathbf{x}^* = \arg \max_{\substack{\mathbf{x} \in \mathbb{Z}_{\geq 0}^{V_s} \\ \|\mathbf{x}\|_1 = k, \mathbf{x} \leq \mathbf{c}}} \sigma(\mathbf{x}).$$

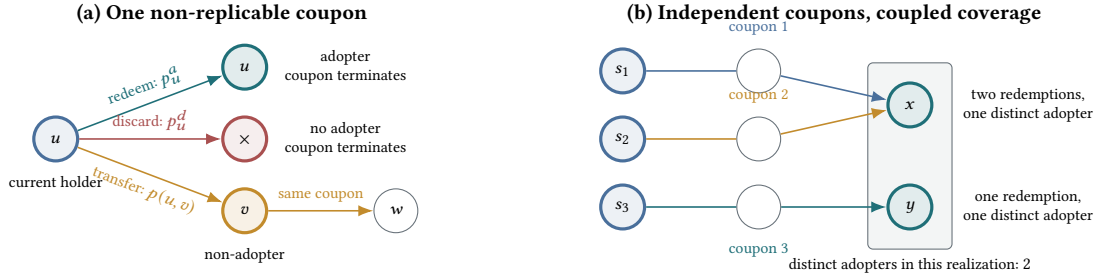


Figure 1: Coupon diffusion and multi-coupon overlap. Redemption and discard terminate a coupon, whereas transfer moves the same coupon to one neighbor. Coupons diffuse independently, but multiple redemptions by the same user contribute only one distinct adopter.

Definition 3.2 (Coupon Usage-Rate Maximization). Given the same input as in Definition 3.1, coupon usage-rate maximization seeks an allocation $\mathbf{x}^\dagger$ that maximizes the expected coupon usage rate:

$$\mathbf{x}^\dagger = \arg \max_{\substack{\mathbf{x} \in \mathbb{Z}_{\geq 0}^{V_S} \\ \|\mathbf{x}\|_1 = k, \mathbf{x} \leq \mathbf{c}}} \eta(\mathbf{x}).$$

In this paper, CIM refers to the adoption-spread maximization problem unless otherwise stated. The usage-rate objective is a complementary objective that is useful when the campaign organizer mainly cares about how many issued coupons are redeemed. Because $\rho(\mathbf{x})$ is linear in $\mathbf{x}$, usage-rate maximization fills seed capacities in descending order of the single-coupon redemption probability $\sum_u q_{vu}$. In contrast, $\sigma(\mathbf{x})$ penalizes overlap at the same adopters and therefore creates a non-trivial allocation problem.

For adoption spread, assigning every coupon to its individually best seed may be suboptimal because several coupons may end at the same adopter, who is counted only once in $\sigma(S)$. Conversely, graph distance alone is insufficient: a distant seed with a low probability of reaching an adopter may contribute little. CIM must therefore balance path-based reachability against overlap among the adopters reached by different coupons. Section 5 uses an ordered tuple only to associate each allocated coupon with an independent RR component; any ordering of the same allocation vector has the same expected spread.

For example, consider two independent realizations: coupon 1 is seeded at s_1 in realization W_1 , and coupon 2 is seeded at s_2 in realization W_2 . In realization W_1 , the forwarding path from s_1 reaches node u , and the adoption event $A_{u,1}$ occurs, so u redeems coupon 1 and becomes an adopter. In realization W_2 , coupon 2 follows a different forwarding path and is redeemed by node v . Node w lies on the forwarding chains but does not redeem either coupon; it helps propagation but does not contribute to the objective. Therefore, in this realization, the number of adopters is two.

4 THEORETICAL ANALYSIS

We first prove that CIM is NP-hard, even on a two-layer diffusion graph. We then show that adoption spread is monotone and has diminishing returns over integer coupon allocations. This structure yields a $(1 - 1/e)$ approximation for exact greedy, but computing

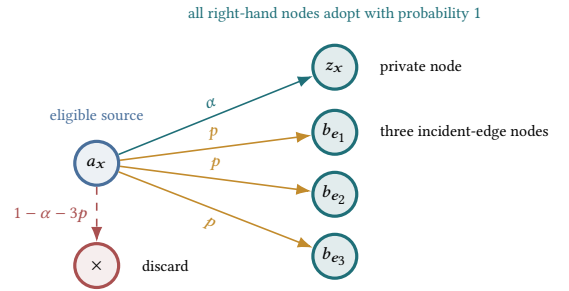


Figure 2: The reduction gadget for a vertex x in the cubic graph. Only a_x is eligible for coupon allocation; a coupon reaches either the private node z_x , one of the three incident-edge nodes, or the discard outcome.

its marginal gains remains expensive; this motivates the reverse-sampling algorithm in Section 5.

4.1 Computational Hardness

The following theorem states the computational hardness of CIM.

THEOREM 4.1. *The CIM problem is NP-hard.*

PROOF. We reduce from vertex cover on cubic graphs, which is NP-hard [5]. Let $H = (V_H, E_H)$ be a cubic graph and let $K \leq |V_H|$. A vertex cover of size at most K can be padded to size exactly K , so we use the exact-size form below. For every vertex $x \in V_H$, create an eligible source node a_x and a private adoption node z_x . For every edge $e \in E_H$, create an edge-adoption node b_e . Only the source nodes are eligible seeds. Set their capacities to K .

Fix constants $\alpha = 1/2$ and $p = 1/24$. A source a_x adopts with probability 0, transfers a coupon to its private node z_x with probability α , and transfers it to each of the three edge nodes b_e incident to x with probability p . The remaining probability $1 - \alpha - 3p$ is assigned to discard. Every z_x and b_e adopts with probability 1. Thus every coupon takes at most one transfer step, and the construction is a valid CIM instance; Figure 2 shows the gadget associated with one vertex x .

We first show that an optimal allocation of K coupons uses K distinct sources, even though CIM permits repeated seeds. Suppose an allocation places $r \geq 2$ coupons at a_x and none at a_y . Removing

one coupon from a_x decreases the adoption probability of z_x by $\alpha(1-\alpha)^{r-1} \leq \alpha(1-\alpha)$ and decreases the total contribution of the three incident edge nodes by at most $3p$. Placing that coupon at a_y increases the adoption probability of its previously uncovered private node z_y by α ; any gain at edge nodes is additional. The net change is therefore at least

$$\alpha - \alpha(1-\alpha) - 3p = \alpha^2 - 3p > 0.$$

Hence a repeated source cannot occur in an optimal allocation.

Consider now a set S of K distinct sources, and let $c_e(S) \in \{0, 1, 2\}$ be the number of selected endpoints of edge e . The private nodes contribute exactly $K\alpha$, while b_e adopts with probability $1 - (1-p)^{c_e(S)}$. Let n_i be the number of edges with $c_e(S) = i$. Since H is cubic,

$$n_0 + n_1 + n_2 = |E_H|, \quad n_1 + 2n_2 = 3K.$$

The total expected adoption spread is

$$\begin{aligned} \sigma(S) &= K\alpha + n_1p + n_2(2p - p^2) \\ &= K\alpha + 3Kp - (3K - |E_H|)p^2 - n_0p^2. \end{aligned}$$

It reaches

$$B = K\alpha + 3Kp - (3K - |E_H|)p^2$$

if and only if $n_0 = 0$, which holds exactly when the corresponding vertices form a size- K vertex cover of H . Because an optimal allocation uses distinct sources, H has a size- K vertex cover if and only if the constructed CIM instance admits an allocation with spread at least B . The construction is polynomial, so CIM is NP-hard. $\square$

4.2 Monotonicity and Submodularity

Because several coupons may be assigned to the same seed, the natural domain of CIM is the integer lattice $\mathbb{Z}_{\geq 0}^{V_s}$. For allocations $\mathbf{x} \leq \mathbf{y}$ componentwise and a candidate seed v , let $\mathbf{e}_v$ denote one additional coupon at v . A function f has the diminishing-returns (DR) property if

$$f(\mathbf{x} + \mathbf{e}_v) - f(\mathbf{x}) \geq f(\mathbf{y} + \mathbf{e}_v) - f(\mathbf{y}).$$

This is the integer-lattice analogue of set-function submodularity.

THEOREM 4.2. *For every coupon influence instance, $\sigma(\mathbf{x})$ is monotone and DR-submodular over coupon allocations.*

PROOF. The marginal gain of assigning one additional coupon to v is

$$\begin{aligned} \Delta_v(\mathbf{x}) &= \sigma(\mathbf{x} + \mathbf{e}_v) - \sigma(\mathbf{x}) \\ &= \sum_{u \in V} q_{vu} \prod_{w \in V_s} (1 - q_{wu})^{x_w}. \end{aligned}$$

Every term is non-negative, so σ is monotone. If $\mathbf{x} \leq \mathbf{y}$, then for every user u ,

$$\prod_{w \in V_s} (1 - q_{wu})^{x_w} \geq \prod_{w \in V_s} (1 - q_{wu})^{y_w}.$$

Therefore $\Delta_v(\mathbf{x}) \geq \Delta_v(\mathbf{y})$, proving the DR property. $\square$

Starting from $\mathbf{x}_0 = \mathbf{0}$, exact greedy assigns coupon $i + 1$ to $v_i \in \arg \max_{v: x_{i,v} < c_v} \Delta_v(\mathbf{x}_i)$ for $i = 0, \dots, k - 1$. Equivalently, we may replace each candidate v by c_v identical copies and apply cardinality-constrained greedy to the resulting ground set. The

standard cardinality argument for monotone DR-submodular functions gives

$$\sigma(\mathbf{x}_k) \geq \left(1 - \left(1 - \frac{1}{k}\right)^k\right) \sigma(\mathbf{x}^*) \geq (1 - 1/e) \sigma(\mathbf{x}^*)$$

[15].

However, an exact greedy implementation is still inefficient, because evaluating $\sigma(S)$ exactly is computationally expensive. One may first estimate the full matrix $Q = (q_{vu})$ using many independent single-coupon trajectories from every candidate seed and then evaluate all greedy marginals, but this requires $O(n^2)$ storage and a large number of forward simulations before seed selection begins. One may also estimate $\sigma(S)$ by Monte Carlo simulation, but a greedy algorithm would require repeated simulations for many candidate seeds and many rounds, leading to a cost of $O(k^2 |V_s| R \bar{L})$, where R is the number of simulations and $\bar{L}$ is the expected coupon-walk length. Therefore, to make the monotone DR-submodular structure algorithmically useful, we need an efficient estimator of the adoption spread. In the next section, we develop such an estimator using k -joint reverse reachable sets, and then build the CIM-RIS algorithm on top of this estimator.

5 ALGORITHM: CIM-RIS

We now design an efficient algorithm for CIM based on reverse influence sampling. We first define reverse reachable sets for single-coupon diffusion and show their coverage property. We then extend this construction to k coupons by introducing k -joint RR sets, which correctly preserve the coupling between the j -th seed and the j -th coupon realization. Finally, we use these samples to estimate the marginal gain of each successive coupon and obtain the CIM-RIS algorithm.

5.1 Reverse Reachable Sets for Coupon Diffusion

The key insight of classical RIS is that influence maximization can be turned into a maximum coverage problem over sampled *reverse reachable (RR) sets*. In this section, we adapt this construction to the coupon influence model. The main difference is that a node becomes an adopter only when two events occur simultaneously: the coupon reaches the node through a live-edge path, and the node takes the adoption action in the corresponding realization.

Definition 5.1 (Live edge). Edge $(w, u) \in E$ is a *live edge* in realization W_j if $\Phi_j(w) = u$.

Single-coupon RR set. Fix a coupon index j and its realization W_j . For a target node u , we define the reverse reachable set $R_j(u, W_j)$ as the set of seed nodes from which coupon j reaches node u and is redeemed there in this realization. Formally,

$$R_j(u, W_j) = \begin{cases} \{w \in V : w \xrightarrow{*} u \text{ in } W_j\}, & \text{if } A_{u,j} = 1, \\ \emptyset, & \text{otherwise,} \end{cases}$$

where $w \xrightarrow{*} u$ denotes a live-edge path from w to u . The condition $A_{u,j} = 1$ is the adoption gate at the root node u . If this gate does not fire, then no seed can cause u to redeem coupon j in realization W_j , and the reverse reachable set is empty. If it does fire, then $R_j(u, W_j)$

contains exactly the nodes that can reach u through live edges in W_j .

LEMMA 5.2 (COVERAGE LEMMA). *Coupon j seeded at s is redeemed by node u in realization W_j if and only if $s \in R_j(u, W_j)$.*

PROOF. $\text{Adopt}_{u,j}(s)$ requires (i) $A_{u,j} = 1$ and (ii) a live-edge path from s to u in W_j . Condition (i) is the adoption gate in the definition of $R_j(u, W_j)$, and condition (ii) is precisely $s \in R_j(u, W_j)$. $\square$

Lemma 5.2 is the single-coupon analogue of the standard RR-set coverage property: testing whether s belongs to $R_j(u, W_j)$ is equivalent to testing whether coupon j seeded at s is redeemed by target node u in realization W_j .

Why a naive union fails for k coupons. For k coupons assigned according to an ordered seed tuple $(s_1, \dots, s_k)$ under independent realizations $(W_1, \dots, W_k)$, node v becomes an adopter if and only if $\bigvee_{j=1}^k \{s_j \in R_j(v, W_j)\}$. A tempting approach is to generate the union $Q(v) = \bigcup_{j=1}^k R_j(v, W_j)$ and check whether any selected seed falls in $Q(v)$. This approach is incorrect because it ignores coupon identities. If $s_i \in R_j(v, W_j)$ for some $i \neq j$, the union test would allow seed s_i to borrow the realization of coupon j , although coupon i and coupon j propagate independently. Therefore, correctness requires a per-coupon test: seed s_j must be checked only against $R_j(v, W_j)$.

k -joint RR sets. Based on the above observation, we define a k -joint RR set rooted at v as the tuple

$$\mathcal{R}(v) = (R_1(v, W_1), \dots, R_k(v, W_k)),$$

where $W_1, \dots, W_k$ are sampled independently. The tuple keeps the correspondence between coupon index j and its own reverse reachable set $R_j(v, W_j)$. For an ordered seed tuple $S = (s_1, \dots, s_k)$, the sample is covered if

$$\bigvee_{j=1}^k \{s_j \in R_j(v, W_j)\}$$

holds. Figure 3 illustrates why this index correspondence is part of the sample rather than a bookkeeping detail. This is the basic estimator used by CIM-RIS.

LEMMA 5.3 (UNBIASEDNESS). *Let v be drawn uniformly from V , and let $(R_1(v, W_1), \dots, R_k(v, W_k))$ be generated from k independent realizations rooted at v . For any ordered seed tuple $S = (s_1, \dots, s_k)$, we have*

$$\sigma(S) = n \cdot \mathbb{E} \left[\mathbb{I} \left[\bigvee_{j=1}^k \{s_j \in R_j(v, W_j)\} \right] \right],$$

or equivalently,

$$\mathbb{E} \left[\mathbb{I} \left[\bigvee_{j=1}^k \{s_j \in R_j(v, W_j)\} \right] \right] = \frac{\sigma(S)}{n}.$$

Algorithm 1 Single-Node Transition Kernel

Require: Node u ; coupon index j ; random number $r_{u,j}$

Require: Parameters $p_u^a, p_u^d, \{p(u, v)\}_{v:(u,v) \in E}$

Ensure: Action $\Phi_j(u) \in \{\text{adopt}, \text{discard}\} \cup N^+(u)$

```

1: if  $r_{u,j} \leq p_u^a$  then
2:   return adopt
3: else if  $r_{u,j} \leq p_u^a + p_u^d$  then
4:   return discard
5: else
6:   Find  $v^*$  such that  $r_{u,j}$  falls in  $v^*$ 's transfer interval
7:   return  $v^*$ 
8: end if
```

PROOF. The expectation is over the uniformly sampled root v and over the independent realizations $W_1, \dots, W_k$.

$$\begin{aligned} \mathbb{E} \left[\mathbb{I} \left[\bigvee_{j=1}^k \{s_j \in R_j(v, W_j)\} \right] \right] &= \frac{1}{n} \sum_{v \in V} \Pr \left[\bigvee_{j=1}^k \{s_j \in R_j(v, W_j)\} \right] \\ &\stackrel{(*)}{=} \frac{1}{n} \sum_{v \in V} \Pr \left[\bigvee_{j=1}^k \text{Adopt}_{u,j}(s_j) \right] \\ &= \frac{\sigma(S)}{n}, \end{aligned}$$

where step $(*)$ applies Lemma 5.2 to each coupon realization. The last equality follows from the definition of adoption spread. $\square$

5.2 Sampling Procedure

The uniform-root construction in Lemma 5.3 is conceptually simple, but it generates an empty joint sample whenever none of the root adoption gates fires. We avoid these zero-contribution samples by conditioning on the event that at least one root-adoption gate fires. For a root v , define

$$C_v = \bigvee_{j=1}^k A_{v,j}, \quad w_v = \Pr[C_v] = 1 - (1 - p_v^a)^k, \quad W = \sum_{v \in V} w_v.$$

If $W = 0$, no node can become an adopter and the optimal spread is zero, so we assume $W > 0$. One conditioned joint sample is generated by (a) drawing root v with probability w_v/W ; (b) drawing the vector of k Bernoulli adoption gates from its product distribution conditioned on C_v ; and (c) for every index j whose gate fires, generating $R_j(v, W_j)$ by a backward traversal over live in-edges. The conditioned gate vector can be drawn in $O(k)$ time by sampling its first firing index from the corresponding truncated geometric distribution and then sampling all later gates independently. Algorithm 1 gives the deterministic transition kernel for one node once the random number $r_{u,j}$ has been fixed. This kernel is used lazily during the backward traversal so that the same node-coupon pair has a consistent action within the same realization. The categorical action distribution of each node can be stored in an alias table using $O(m+n)$ total preprocessing, allowing each action to be generated and cached in $O(1)$ time while preserving the same distribution.

Algorithm 2 implements the conditioned procedure. For each coupon index j with $A_{v,j} = 1$, it performs a backward BFS. When the traversal examines an in-neighbor w of the current node, it

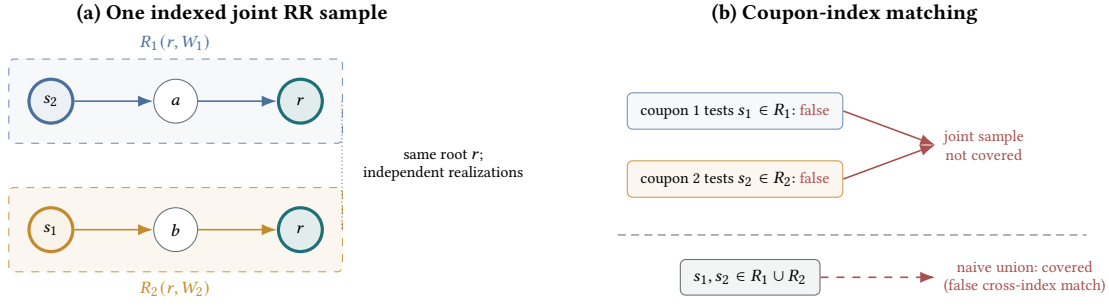


Figure 3: A k -joint RR sample preserves the correspondence between coupon index j and realization W_j (shown for $k = 2$). Coverage must test $s_j \in R_j$ with matching indices; taking the union of the RR sets can create a false cross-index match.

Algorithm 2 Conditioned k -Joint RR Sample Generation

Require: Graph $G = (V, E)$; parameters $\{p_u^a, p_u^d, p(u, v)\}$; coupon count k ; root weights $\{w_v\}$ and W

Ensure: k -joint RR sample $((R_1, \dots, R_k), v)$

```

1: Sample root  $v$  with probability  $w_v/W$ 
2: Sample  $(A_{v,1}, \dots, A_{v,k})$  from their product Bernoulli( $p_v^a$ ) distribution conditioned on  $C_v$ 
3:  $R_1, \dots, R_k \leftarrow \emptyset$ 
4: for  $j = 1$  to  $k$  do
5:   if  $A_{v,j} = 1$  then
6:      $R_j \leftarrow \{v\}$ ;  $\text{queue} \leftarrow [v]$ 
7:     while  $\text{queue} \neq \emptyset$  do
8:        $u \leftarrow \text{queue.dequeue}()$ 
9:       for each in-neighbor  $w$  of  $u$  do
10:        if  $w \notin R_j$  then
11:          if  $r_{w,j}$  not yet sampled then
12:             $r_{w,j} \sim \text{Uniform}[0, 1]$ 
13:          end if
14:          if  $\Phi_j(w) = u$  then
15:             $R_j \leftarrow R_j \cup \{w\}$ ;  $\text{queue.enqueue}(w)$ 
16:          end if
17:        end if
18:      end for
19:    end while
20:  end if
21: end for
22: return  $((R_1, \dots, R_k), v)$ 

```

generates $r_{w,j}$ only if this random number has not been generated before, and then checks whether (w, u) is the live edge selected by w in realization W_j . If so, w is added to R_j ; otherwise, this in-neighbor is ignored.

LEMMA 5.4 (CONDITIONED-ROOT ESTIMATOR). *Let $\mathcal{R}$ be generated by Algorithm 2, and let*

$$Y(S_i) = \mathbb{I}\left[\bigvee_{j=1}^i \{s_j \in R_j\}\right]$$

for any prefix $S_i = (s_1, \dots, s_i)$ with $i \leq k$. Then

$$\mathbb{E}[WY(S_i)] = \sigma(S_i),$$

where $\sigma(S_i)$ is the spread produced by the i coupons in the prefix.

PROOF. For a fixed root v with $w_v > 0$, let $Y_v(S_i)$ be the coverage event under the unconditioned realizations. Coverage by any prefix implies that at least one of the first i root adoption gates fires. It therefore also implies the conditioning event over all k gates, so $Y_v(S_i) \subseteq C_v$ and $\Pr[Y_v(S_i) | C_v] = \Pr[Y_v(S_i)]/w_v$. Taking expectation over the importance-sampled root gives

$$\begin{aligned} \mathbb{E}[WY(S_i)] &= \sum_{v \in V} \frac{w_v}{W} W \Pr[Y_v(S_i) | C_v] \\ &= \sum_{v \in V} \Pr[Y_v(S_i)] = \sigma(S_i), \end{aligned}$$

where the final equality follows from Lemma 5.2 and the definition of adoption spread. $\square$

5.3 The CIM-RIS Algorithm

Let $\mathcal{F}$ be a collection of T independently generated k -joint RR samples,

$$\mathcal{F} = \{((R_1^{(t)}, \dots, R_k^{(t)}), v^{(t)})\}_{t=1}^T.$$

For a prefix $S_i = (s_1, \dots, s_i)$ and sample t , define

$$Y_t(S_i) = \mathbb{I}\left[\bigvee_{j=1}^i \{s_j \in R_j^{(t)}\}\right].$$

By Lemma 5.4, $\hat{\sigma}(S_i) = W \sum_t Y_t(S_i)/T$ is an unbiased spread estimator. At round i , assigning coupon i to u newly covers sample t exactly when $Y_t(S_{i-1}) = 0$ and $u \in R_i^{(t)}$. We therefore estimate its marginal gain by

$$\hat{\Delta}_i(u | S_{i-1}) = \frac{W}{T} \sum_{t=1}^T \mathbb{I}[Y_t(S_{i-1}) = 0] \mathbb{I}[u \in R_i^{(t)}].$$

LEMMA 5.5 (MARGINAL-GAIN ESTIMATOR). *For every fixed prefix S_{i-1} and feasible candidate $u \in V_s$ with $x_u(S_{i-1}) < c_u$,*

$$\mathbb{E}[\hat{\Delta}_i(u | S_{i-1})] = \sigma(x(S_{i-1}) + e_u) - \sigma(x(S_{i-1})).$$

PROOF. For a fixed root v , the first indicator means that none of the first $i-1$ coupons is redeemed at v , while the second means that coupon i seeded at u is redeemed there. Their conjunction is exactly the event that the additional coupon makes v a new adopter. This event implies the conditioning event C_v even though C_v is

defined using all k coupon gates. The same importance-reweighting calculation as in Lemma 5.4 therefore gives

$$\mathbb{E}\left[W\mathbb{I}[Y_t(S_{t-1}) = 0]\mathbb{I}[u \in R_i^{(t)}]\right] = \sum_{v \in V} \Pr[v \text{ is newly adopted}].$$

The sum on the right is the stated marginal gain. $\square$

Let $N = |V_s|$, let $\text{OPT} = \sigma(\mathbf{x}^*)$, and let $\epsilon, \delta \in (0, 1)$ and $\text{LB} \in (0, \text{OPT}]$ be a valid lower bound. We set

$$T = \left\lceil \frac{16k^2W}{\epsilon^2\text{LB}^2} \left(k \ln N + \ln \frac{2k}{\delta} \right) \right\rceil.$$

A deterministic choice is

$$\text{LB}_{\text{path}} = \max_{\substack{s=v_0 \in V_s, v_\ell \in V \\ (v_0, \dots, v_\ell) \text{ simple}}} p_{v_\ell}^a \prod_{h=0}^{\ell-1} p(v_h, v_{h+1}),$$

the largest probability of following one seed-to-adopter path. If this value is zero, then $\text{OPT} = 0$; otherwise it is a positive lower bound on OPT . Because all path factors lie in $[0, 1]$, a maximizing positive path can be chosen simple; the value can be computed using shortest paths after applying negative logarithms to positive transfer and adoption probabilities. Independent pilot RR samples can give a tighter lower confidence bound in practice. The value of T above is a sufficient theorem-level sample count: when ϵ, δ , and a valid LB are supplied, it is the quantity used to attach the high-probability guarantee in Theorem 5.6 to one run. Because the bound is uniform over all possible greedy prefixes, it can be conservative in practice, particularly when LB is small. An implementation may instead use a fixed empirical sample budget, but such a run should be interpreted as an approximate RIS implementation without a per-run (ϵ, δ) certificate unless its budget also satisfies this bound. With these samples, CIM-RIS processes coupon indices in order and selects the candidate with the largest empirical marginal gain. Using inverted lists from node-coupon pairs to samples, the gains in each round are computed by scanning RR memberships rather than all candidate-sample pairs.

5.4 Theoretical Guarantees

THEOREM 5.6. *Let $\text{LB} \in (0, \text{OPT}]$ and let T be chosen as above. CIM-RIS returns an allocation $\hat{\mathbf{x}}$ satisfying, with probability at least $1 - \delta$,*

$$\sigma(\hat{\mathbf{x}}) \geq (1 - 1/e - \epsilon) \text{OPT}.$$

Its expected running time is

$$O\left(\frac{k^3(m+n)}{\epsilon^2\text{LB}^2} \left(k \ln |V_s| + \ln \frac{k}{\delta}\right) + k|V_s|\right).$$

PROOF. *Approximation.* Lemma 5.5 shows that every coverage count used by the algorithm is an unbiased estimate of the corresponding true marginal gain. There are at most $\sum_{i=1}^k |V_s|^i \leq k|V_s|^k$ prefix-candidate combinations over all possible executions. For a fixed combination, define

$$Z_t = W\mathbb{I}[Y_t(S_{t-1}) = 0]\mathbb{I}[u \in R_i^{(t)}].$$

Then $0 \leq Z_t \leq W$, $\mathbb{E}[Z_t] = \Delta_u(\mathbf{x}(S_{t-1}))$, and this marginal is at most one because one additional coupon can create at most one

Algorithm 3 CIM-RIS($G, V_s, c, k, \epsilon, \delta, \text{LB}$)

```

1: Compute  $w_v \leftarrow 1 - (1 - p_v^a)^k$  for all  $v$  and  $W \leftarrow \sum_v w_v$ 
2: if  $W = 0$  then
3:   return any feasible allocation with spread 0
4: end if
5: Set  $T$  according to the sample-size equation above
6: Generate  $T$  independent conditioned samples  $\mathcal{F}$  using Algorithm 2
7:  $S_0 \leftarrow ()$ ;  $\mathbf{x} \leftarrow \mathbf{0}$ ; covered[ $t$ ]  $\leftarrow$  false for every  $t$ 
8: for  $i = 1$  to  $k$  do
9:   for  $u \in V_s$  with  $x_u < c_u$  do
10:     $g(u) \leftarrow |\{t : \neg \text{covered}[t] \wedge u \in R_i^{(t)}\}|$ 
11:   end for
12:    $s_i \leftarrow \arg \max_{u \in V_s: x_u < c_u} g(u)$ ;  $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{e}_{s_i}$ 
13:   for each  $t$  with  $\neg \text{covered}[t]$  and  $s_i \in R_i^{(t)}$  do
14:     covered[ $t$ ]  $\leftarrow$  true
15:   end for
16:    $S_i \leftarrow (S_{i-1}, s_i)$ 
17: end for
18: return allocation  $\mathbf{x}$  and ordered tuple  $S_k$ 

```

new adopter. Consequently, $\text{Var}(Z_t) \leq \mathbb{E}[Z_t^2] \leq W\mathbb{E}[Z_t] \leq W$. Bernstein's inequality therefore gives

$$\Pr\left[|\widehat{\Delta}_i(u | S_{i-1}) - \Delta_u(\mathbf{x}(S_{i-1}))| > \tau\right] \leq 2 \exp\left(-\frac{T\tau^2}{2W + (2/3)W\tau}\right).$$

Set $\tau = \epsilon\text{LB}/(2k)$. Since $\text{LB} \leq \text{OPT} \leq k$, we have $\tau \leq 1/2$. Let $L = k \ln |V_s| + \ln(2k/\delta)$. The denominator in the Bernstein exponent is at most $7W/3$, whereas $T\tau^2 \geq 4WL$. Thus the failure probability for one combination is at most $2e^{-L} = \delta/(k|V_s|^k)$. A union bound makes every marginal estimate accurate within τ simultaneously with probability at least $1 - \delta$. On this event, the candidate selected in each round has true marginal gain at least the largest feasible marginal minus $2\tau = \epsilon\text{LB}/k$.

Let $\mathbf{x}_i$ be the allocation after round i , and let $\mathbf{x}_i \vee \mathbf{x}^*$ denote the componentwise maximum of $\mathbf{x}_i$ and an optimal allocation. Monotonicity and DR-submodularity give

$$\text{OPT} - \sigma(\mathbf{x}_i) \leq \sigma(\mathbf{x}_i \vee \mathbf{x}^*) - \sigma(\mathbf{x}_i) \leq \sum_{v \in V_s} (x_v^* - x_{i,v})_+ \Delta_v(\mathbf{x}_i).$$

Every term with $(x_v^* - x_{i,v})_+ > 0$ is feasible because $x_{i,v} < x_v^* \leq c_v$, and the coefficients sum to at most k . Therefore,

$$\max_{v \in V_s: x_{i,v} < c_v} \Delta_v(\mathbf{x}_i) \geq \frac{\text{OPT} - \sigma(\mathbf{x}_i)}{k}.$$

Combining this inequality with the concentration event gives

$$\text{OPT} - \sigma(\mathbf{x}_{i+1}) \leq \left(1 - \frac{1}{k}\right) (\text{OPT} - \sigma(\mathbf{x}_i)) + \frac{\epsilon\text{LB}}{k}.$$

Writing $g_i = \text{OPT} - \sigma(\mathbf{x}_i)$ and iterating this recurrence for k rounds yields

$$\begin{aligned} g_k &\leq \left(1 - \frac{1}{k}\right)^k \text{OPT} + \frac{\epsilon\text{LB}}{k} \sum_{r=0}^{k-1} \left(1 - \frac{1}{k}\right)^r \\ &\leq e^{-1} \text{OPT} + \epsilon\text{LB} \leq (e^{-1} + \epsilon) \text{OPT}. \end{aligned}$$

Hence $\sigma(\mathbf{x}_k) \geq (1 - 1/e - \epsilon) \text{OPT}$.

Complexity. Let q_{wv} be the probability that one coupon seeded at w is redeemed by v . For coupon index j , the event $w \in R_j(v, W_j)$ implies C_v , and hence, for every root with $w_v > 0$,

$$\Pr[w \in R_j(v, W_j) \mid C_v] = \frac{q_{wv}}{w_v}.$$

After averaging over the root distribution w_v/W , the expected total number of node memberships in one conditioned joint sample is

$$\frac{k}{W} \sum_{w \in V} \sum_{v \in V} q_{wv} \leq \frac{kn}{W}.$$

The inequality holds because a coupon seeded at w can be redeemed by at most one node, so $\sum_v q_{wv} \leq 1$. The corresponding expected number of scanned in-edges is

$$\frac{k}{W} \sum_{w \in V} d^-(w) \sum_{v \in V} q_{wv} \leq \frac{km}{W}.$$

Sampling the conditioned root gates takes $O(k)$ time, which is absorbed because $W \leq n$. Alias tables for the root and node-action distributions take $O(m+n)$ total preprocessing time; afterward, each root and each newly encountered node action can be sampled in $O(1)$ time. Thus one conditioned joint sample costs $O(k(m+n)/W)$ in expectation. Multiplying by T cancels W and gives the first term in the stated bound. The ceiling in T and the $O(m+n)$ preprocessing do not change this order. Indeed, $\text{OPT} \leq k$ because each coupon creates at most one adopter, and $\text{OPT} \leq W$ because adoption at root v implies C_v ; hence $\text{LB} \leq \min\{k, W\}$. These inequalities ensure that the stated sampling term also dominates the cost of one additional sample from the ceiling and the preprocessing cost. Finally, inverted RR-membership lists account for the sample-processing work, while initializing candidate gains in each round costs $O(k|V_s|)$. $\square$

Compared with classical RIS, the additional dependence on k is caused by independent coupon realizations and sequential marginal estimation. The explicit dependence on LB reflects the fact that adoption probabilities can be arbitrarily small: unlike classical IM, a seed user is not automatically counted as an adopter.

6 EXPERIMENTS

Our evaluation addresses four questions: (Q1) how closely does CIM-RIS approach Monte Carlo greedy; (Q2) how do broadcast-IM and structural baselines behave under coupon diffusion; (Q3) when does conditioned root sampling improve estimator efficiency; and (Q4) how does running time scale with the coupon budget and graph size? We use five graph instances derived from real networks and three parameter regimes that balance adoption, discard, and forwarding differently.

6.1 Experimental Setup

Datasets. We use five graph instances derived from real networks in the KONECT collection [9], with basic statistics shown in Table 1. The Netscience dataset is a scientific collaboration network, EmailEnron is an email communication network, and NetFacebookEgo and Douban are online social networks. DoubanRandom is a sampled Douban instance. We represent each undirected contact by two directed edges. The four smaller graphs form the primary quality suite, while network.douban is reserved for scalability tests.

Table 1: Datasets used in the experiments.

Dataset	$ V $	$ E $	Avg. deg.	Max outdeg.	Sparsity
Netscience	379	1,828	4.82	34	98.72%
NetFacebookEgo	2,888	5,962	2.06	769	99.93%
DoubanRandom	4,723	11,774	2.49	45	99.95%
EmailEnron	33,696	361,622	10.73	1,383	99.97%
network.douban	154,907	654,206	4.22	287	99.97%

Algorithm 4 Degree-Based Node Probability Generation

Require: Degrees $\{d_u\}$; parameters $(base_a, slope_a, base_d, slope_d, h)$
Ensure: $\{p_u^a, p_u^d, p_u^t\}$ for all u
1: **for** $u \in V$ **do**
2: $\hat{d}_u \leftarrow \lceil \min\{1, \max\{10^{-5}, \log(1+d_u)/\max_w \log(1+d_w)\}\} \rceil^h$
3: **end for**
4: **for** $u \in V$ **do**
5: $p_u^a \leftarrow base_a + slope_a \cdot (1 - \hat{d}_u)$ $\triangleright$ Low-degree nodes adopt more
6: $p_u^d \leftarrow base_d + slope_d \cdot \hat{d}_u$ $\triangleright$ High-degree nodes discard more
7: $p_u^t \leftarrow 1 - p_u^a - p_u^d$
8: **if** $d_u = 0$ **then**
9: $p_u^d \leftarrow p_u^d + p_u^t$; $p_u^t \leftarrow 0$
10: **end if**
11: **end for**
12: $p(u, v) \leftarrow p_u^t/d_u$ for every node with $d_u > 0$
13: **return** $\{(p_u^a, p_u^d, p_u^t)\}_{u \in V}$

Table 2: Parameter settings for the three diffusion scenarios.

Scenario	$base_a$	$slope_a$	$base_d$	$slope_d$
I. Balanced	0.30	0.10	0.10	0.10
II. Adoption-heavy	0.45	0.35	0.15	0.05
III. Forwarding-heavy	0.01	0.05	0.01	0.05

Diffusion parameters. Node probabilities are generated by the degree-based log-continuous procedure in Algorithm 4. This procedure creates heterogeneous behavior in which high-degree hubs adopt less and discard more; their transfer probabilities depend on the scenario parameters. The normalized log-degree $\hat{d}_u$ controls how p_u^a and p_u^d scale with degree.

We evaluate three scenarios reflecting distinct real-world campaign settings, as summarized in Table 2:

- **Scenario I (Balanced):** A balanced campaign in which adoption and forwarding coexist at moderate rates; the parameterization gives $p_u^t = 0.50$ for every node.
- **Scenario II (Adoption-heavy):** Adoption probabilities range from 0.45 to 0.80, making immediate redemption more important than in Scenario I.
- **Scenario III (Forwarding-heavy):** The parameterization gives $p_u^t = 0.93$ for every node, so coupons trace substantially longer forwarding paths. Adoption is rare at each step, but coupons travel farther, making multi-hop graph structure more relevant.

These degree-conditioned probabilities are controlled stress-test parameters rather than estimates from observed coupon campaigns. Accordingly, the experiments establish behavior under the stated model on real network topologies, not external validity for a particular commercial deployment.

Baselines. CIM-RIS is compared against two groups of methods. *Coupon-aware and structural heuristics.*

- **MC-Greedy:** Estimates the single-coupon adoption probabilities q_{vu} from independent forward trajectories and applies exact greedy to the resulting estimated objective. We use 100,000 trajectories per source on Netscience and 50,000 per source on NetFacebookEgo in every scenario. Reference construction and final forward evaluation use independent random streams. Because constructing the full estimated matrix is expensive, this quality reference is limited to these two graphs.
- **1Hop-Sort:** Ranks nodes by expected one-step adoption score $p_u^a + \sum_{v \in N^+(u)} p(u, v) p_v^a$ and selects top- k .
- **Alpha-Sort:** Selects top- k nodes by p_u^a .
- **DegreeTopM:** Selects top- k nodes by out-degree.
- **PageRank:** Selects top- k nodes by PageRank score [17].
- **Random:** k seeds drawn uniformly at random; lower-bound reference.

IC-model adaptation (model-mismatch baseline).

- **IC-RIS:** Generates standard RR sets under the IC model with edge probabilities $p(u, v)$ and applies greedy maximum coverage using the same sample budget as CIM-RIS. This baseline follows the core RIS construction used by scalable IM methods [1, 21] while isolating the effect of replacing single-path coupon diffusion with broadcast diffusion.

The IC baseline tests whether broadcast-oriented reverse sampling can compensate for model mismatch.

Evaluation metrics. For the main comparison, we set $V_s = V$ and $c_v = 1$ for all methods, matching the conventional distinct-seed policy used by the IM baselines. CIM-RIS uses the adoption-root importance sampler described in Section 5; no path-success surrogate is used. Each seed allocation is evaluated by averaging two quantities over 10,000 independent Monte Carlo simulations of the k -coupon influence process: (a) *adoption spread*: distinct users who redeem at least one coupon (primary objective); and (b) *total redemptions*: redeemed coupons, including multiple redemptions by the same user (secondary metric). We use $k \in \{10, 25, 50, 100, 150, 200\}$.

Implementation and reproducibility. All methods are implemented in Python 3.8 and evaluated on a machine with eight AMD EPYC 9654 cores and 16 GB of memory; the reported selection times use one process. The NetFacebookEgo MC-Greedy matrix is generated in 32 chunks using eight processes; this preprocessing time is not included in method selection times. We use five independent selection seeds (20260715–20260719). The joint-sample budget is fixed before the cross-method comparison: 100,000 samples for $k \leq 50$ and 50,000 samples for $k \geq 100$. IC-RIS uses the same sample budget as CIM-RIS. This two-level schedule was fixed from a pre-comparison convergence pilot. It limits total sampling work because one joint sample contains up to k RR components and therefore becomes more expensive as the coupon budget grows;

Table 3: Mean relative adoption-spread difference (%) between CIM-RIS and each baseline. Positive values favor CIM-RIS. Baseline comparisons average 24 graph-budget pairs per scenario; MC-Greedy averages the six Netscience budgets.

Scenario	MC-Greedy	1Hop-Sort	IC-RIS	DegreeTopM	PageRank
Balanced	−2.3	+5.8	−0.1	+12.6	+13.3
Adoption-heavy	−2.3	−0.6	+1.4	+9.5	+8.9
Forwarding-heavy	−3.4	+8.5	−0.7	+25.5	+30.2

the change at $k = 100$ is not an adaptive stopping decision based on the reported method outcomes. These fixed budgets support a controlled empirical comparison; they are not presented as the generally conservative sample count in Theorem 5.6 for a particular (ϵ, δ) pair. Every selected allocation is evaluated with 10,000 fresh forward realizations, and methods within one configuration use common random-number streams. Figure error bars for CIM-RIS show 95% confidence intervals over the five selection runs. Across the 72 graph-scenario-budget configurations, the adoption spread has a mean cross-seed coefficient of variation of 0.51% and a maximum of 3.07%; the mean relative range is 1.27%.

6.2 Solution Quality

Table 3 summarizes the mean relative difference between CIM-RIS and each baseline across the three scenarios. For each graph-budget pair, the relative difference is $100(\bar{\sigma}_{\text{CIM-RIS}} - \bar{\sigma}_{\text{base}})/\bar{\sigma}_{\text{base}}$. It complements the budget-level results in Figure 4 and reports the adoption-heavy results without adding another four-panel figure.

The upper row of Figure 4 reports budget-level adoption spread in the balanced scenario. On Netscience, CIM-RIS is within 1.1–3.8% of the MC-Greedy reference over all six budgets, with a mean gap of 2.3%. Across the four quality graphs, CIM-RIS outperforms 1Hop-Sort at 23 of 24 graph-budget pairs, with a mean relative improvement of 5.8%. It also outperforms DegreeTopM and PageRank at every pair, by 12.6% and 13.3% on average, respectively. IC-RIS is effectively tied in this scenario: each method wins 12 of the 24 graph-budget comparisons and the mean difference is 0.1% in favor of IC-RIS. This result is consistent with the balanced parameters, under which immediate adoption is common and local scores already provide useful seed information. The secondary redemption counts also confirm that the distinct-adopter objective is not interchangeable with coupon utilization. For example, at $k = 200$ on NetFacebookEgo, CIM-RIS produces an average of 128.8 redemptions but only 105.4 distinct adopters; duplicate redemptions account for 18.1% of the redeemed coupons.

The lower row of Figure 4 shows the forwarding-heavy stress test. Absolute spread is lower because per-visit adoption is rare. The mean gap from MC-Greedy on Netscience is 3.4% and decreases to 1.5% at $k = 200$. Across all graph-budget pairs, CIM-RIS outperforms 1Hop-Sort at 20 of 24 points and by 8.5% on average. Its mean improvements over DegreeTopM and PageRank increase to 25.5% and 30.2%, respectively, and it wins all 24 comparisons against each. IC-RIS remains competitive and is 0.7% higher on average, showing that model fidelity does not imply uniform empirical dominance at a fixed sample budget. The adoption-heavy scenario gives an

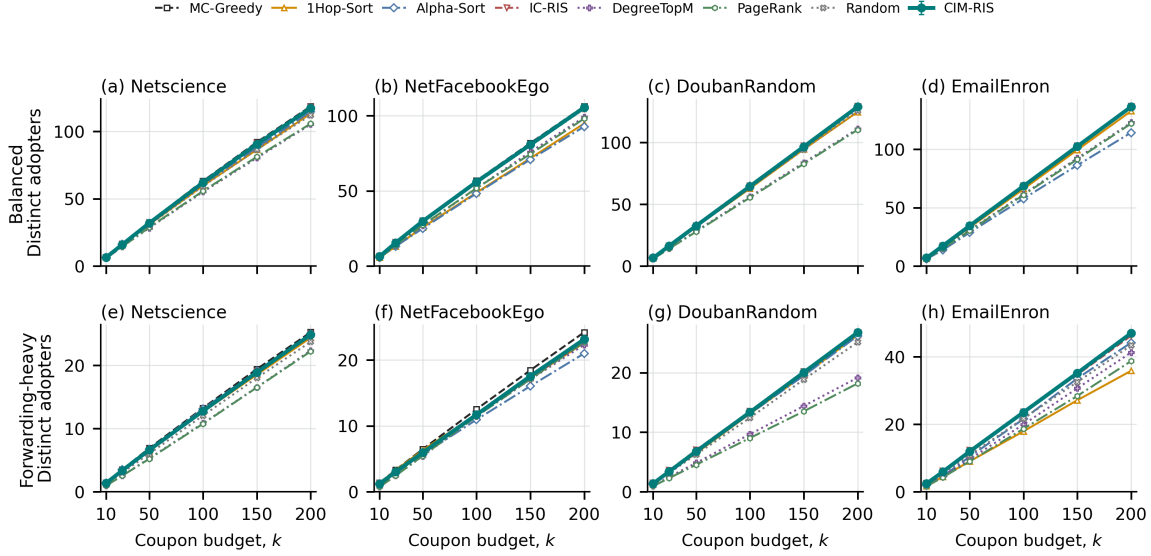


Figure 4: Adoption spread in the balanced (top) and forwarding-heavy (bottom) scenarios. Each point is the mean of five selection runs, each evaluated using 10,000 forward realizations; CIM-RIS error bars show 95% confidence intervals across runs. MC-Greedy is reported on Netscience and NetFacebookEgo.

intermediate pattern: the mean Netscience gap to MC-Greedy is 2.3%, while CIM-RIS outperforms IC-RIS at 23 of 24 points and by 1.4% on average. It again beats both topology-only rankings at all 24 points, although 1Hop-Sort is 0.6% higher on average when immediate adoption dominates.

6.3 Reference Strength, Sample Sensitivity, and Capacity

We extend the strong-reference comparison from Netscience to NetFacebookEgo using 50,000 independent trajectories per source. Figure 5(a) shows that the mean gaps are 1.37% in the balanced scenario and 0.38% in the adoption-heavy scenario. The forwarding-heavy case is harder: the gap decreases from 11.1% at $k = 10$ to 4.4% at $k = 200$, with a mean of 7.2%. Across the two reference graphs, all three scenarios, and all six budgets, the observed gap ranges from 0.3% to 11.1%. These results strengthen the comparison beyond the small Netscience graph, but also identify low-budget, low-adoption diffusion as the least favorable regime for the fixed empirical sample budget.

Figures 5(b) and 5(c) vary the number of joint RR samples in the forwarding-heavy scenario while holding the evaluation streams fixed. For every combination of the two graphs and $k \in \{10, 50, 200\}$, increasing the sample count from 5,000 to 100,000 reduces the mean gap to MC-Greedy. Averaged over the six combinations, the gap decreases from 8.90% to 5.09%. The improvement is largest on Netscience at $k = 10$, where the gap falls from 10.72% to 3.61%, while NetFacebookEgo at $k = 10$ remains the hardest case, decreasing from 14.29% to 11.00%. Thus the sampling budget explains part, but not all, of the remaining difference from MC-Greedy. The corresponding running times expose the expected cost–accuracy trade-off. Averaged over the six network–budget combinations, selection

time grows from 0.266 seconds with 5,000 samples to 1.016, 2.435, and 4.741 seconds with 20,000, 50,000, and 100,000 samples, respectively. Over the same budgets, the mean gap decreases from 8.90% to 7.10%, 6.22%, and 5.09%. Runtime is therefore approximately proportional to the number of samples, whereas the observed quality improvement has diminishing returns. This trade-off motivates the fixed two-level schedule in the main comparison, but does not turn that empirical schedule into the theorem-calibrated value of T .

Finally, Figure 5(d) evaluates the capacity formulation at $k = 200$ under $c_v = 1$, $c_v = 2$, and $c_v = k$. In the balanced and adoption-heavy scenarios, both CIM-RIS and MC-Greedy almost always choose distinct seeds even when repetition is allowed, so relaxing capacity has no material effect. In the forwarding-heavy scenario, CIM-RIS uses an average of 49.6–63.4 repeated placements on Netscience and improves spread by 1.60–2.09% over $c_v = 1$; on NetFacebookEgo, it uses 6.8–7.8 repeated placements and improves spread by about 0.25%. Repeated placement is therefore useful when long trajectories make a small group of sources repeatedly valuable, but it is not uniformly beneficial across diffusion regimes.

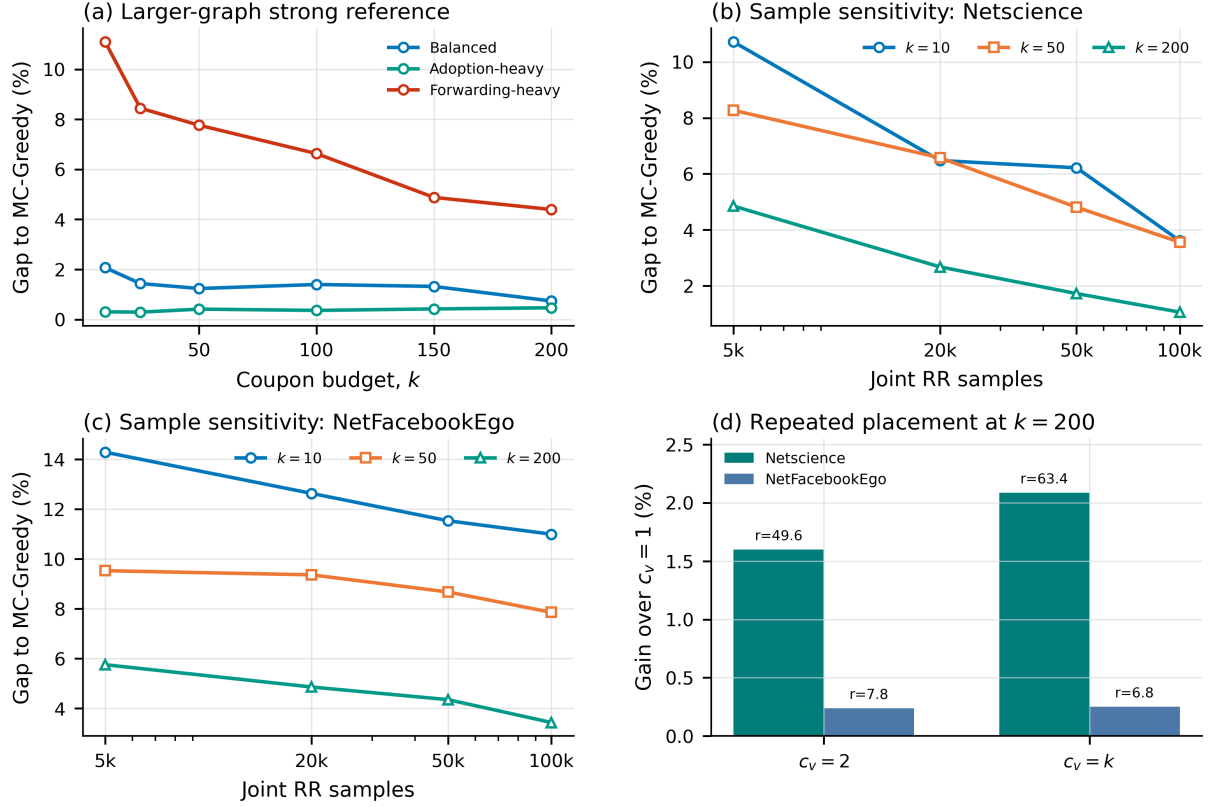


Figure 5: Strong-reference and robustness results. (a) Gap to MC-Greedy on NetFacebookEgo. (b–c) Effect of the joint RR sample count in the forwarding-heavy scenario. (d) Spread gain from relaxing the per-node capacity at $k = 200$; labels report the mean number of repeated placements. Every reported statistic aggregates five CIM-RIS selection seeds, with 10,000 forward realizations per evaluated allocation.

6.4 Conditioned Sampling and Scalability

Figure 6 isolates the conditioned-root sampler in the forwarding-heavy scenario. At $k = 10$, uniform-root sampling spends 68.4% of its Netscience samples and 61.0% of its EmailEnron samples on realizations in which no root adoption gate fires. Conditioning removes this work and reduces the coefficient of variation from 0.128 to 0.087 on Netscience and from 0.809 to 0.492 on EmailEnron. After accounting for sampling time, the corresponding variance-time improvements are 1.30 $\times$ and 1.53 $\times$. Across all six measured conditions, the observed variance-time ratio is between 1.30 $\times$ and 1.59 $\times$ in favor of conditioning. By $k = 200$, however, fewer than 0.3% of uniform-root samples have no firing gate, so differences in empirical variance should be interpreted with the finite 30-batch sample size in mind.

Figure 7 reports end-to-end CIM-RIS selection time, including sample generation and greedy coverage. The drop from $k = 50$ to $k = 100$ on several graphs is caused by the pre-specified halving of the sample budget rather than a faster per-sample computation. At $k = 200$, mean selection time is 51.9 seconds on EmailEnron and 31.2 seconds on the larger network.douban graph. The non-monotone ordering across real graphs reflects different degree distributions

and RR-set sizes; the result should therefore be read as practical scalability evidence rather than a direct empirical fit of the worst-case bound.

7 CONCLUSION

We formalized coupon influence maximization under non-replicable, single-path diffusion. In this model, a coupon is redeemed, discarded, or transferred to one neighbor, and only redemption creates an adopter. The allocation formulation supports both distinct seeds and repeated placements through per-node capacities while counting each adopter once. We proved that CIM is NP-hard and that adoption spread is monotone and DR-submodular over coupon allocations. We then developed k -joint RR samples whose coverage gives unbiased spread and marginal-gain estimates. The resulting CIM-RIS algorithm achieves a $(1 - 1/e - \epsilon)$ approximation with high probability and has expected running time near-linear in the graph size when the budget, accuracy, and spread lower bound are fixed. Experiments show gaps of 0.3–11.1% from high-precision Monte Carlo greedy across two reference graphs and three scenarios; the largest gap occurs for low-budget forwarding-heavy diffusion on NetFacebookEgo and decreases as the RR sample budget grows.

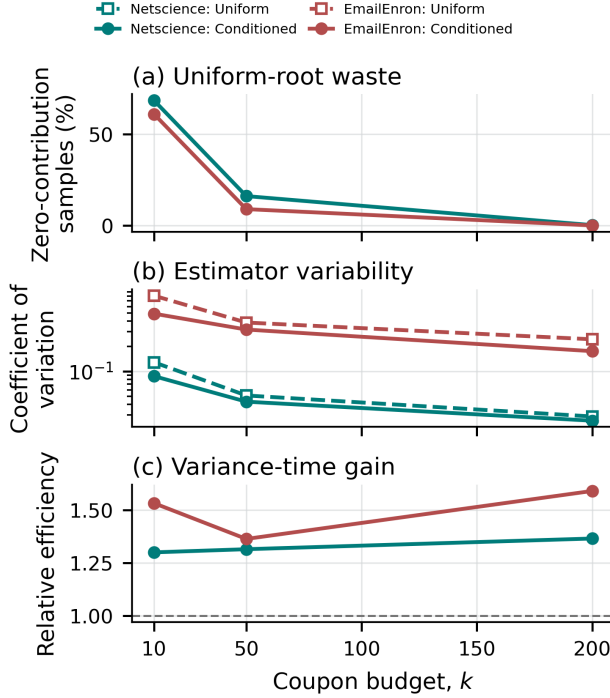


Figure 6: Uniform-root versus conditioned-root estimation in the forwarding-heavy scenario. Each statistic uses 30 independent batches of 20,000 samples. Relative efficiency is the ratio of the variance–time products of uniform and conditioned sampling; values above one favor conditioning.

Across all three scenarios, CIM-RIS outperforms DegreeTopM and PageRank at every one of the 72 graph–budget pairs, with its largest average gains appearing in the forwarding-heavy scenario. Local and broadcast baselines remain competitive on some instances, so no method dominates every graph and budget. Conditioned root sampling removes zero-contribution samples and improves the measured variance–time product by 1.3–1.5 $\times$ at $k = 10$ on the two tested graphs. Capacity experiments further show that repeated placement provides a modest gain in the forwarding-heavy regime but is rarely selected when adoption is common.

Several directions remain open. First, extending the model to *adaptive* allocation, where later placements depend on observed early adoptions, could reduce budget waste in time-sensitive campaigns. Second, coupled campaign rules, such as one redemption per account or forwarding incentives after adoption, require RR samples that encode arrival order and user state. Third, tighter lower-bound estimation and variance reduction for very small adoption probabilities may further improve the practical cost of CIM-RIS.

A ADDITIONAL PARAMETER-REGIME EXPERIMENTS

The three scenarios in the main evaluation provide representative operating points but do not show how the relative performance

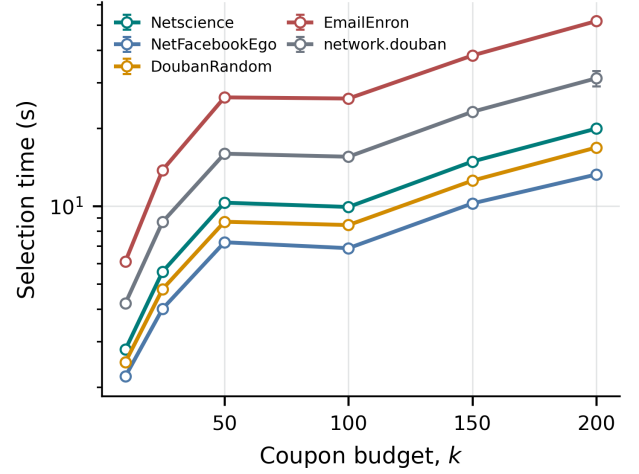


Figure 7: Single-process CIM-RIS selection time in the balanced scenario, averaged over five runs. The sample budget is 100,000 for $k \leq 50$ and 50,000 for $k \geq 100$; error bars show one standard deviation.

changes between them. We therefore conduct a prespecified two-dimensional parameter sweep on Netscience and NetFacebookEgo. This experiment is intended to map favorable and unfavorable regimes rather than to establish that CIM-RIS dominates every alternative.

Let $\hat{d}_u \in [0, 1]$ be the normalized log-degree of node u . For a central redemption share r and transfer probability t , we set

$$h_u = \text{clip}\left(r + 0.20(0.5 - \hat{d}_u), 0.05, 0.95\right),$$

$$p_u^a = (1 - t)h_u, \quad p_u^d = (1 - t)(1 - h_u), \quad p_u^t = t.$$

Thus h_u controls the probability of redemption conditional on stopping, while t controls the expected persistence of a coupon walk. Transfer mass is reassigned to discard at isolated nodes. The full grid uses $t \in \{0.30, 0.50, 0.70, 0.85, 0.93\}$ and $r \in \{0.20, 0.40, 0.60, 0.80\}$ at $k = 100$. We additionally fix $r = 0.60$ and evaluate $k \in \{25, 100, 200\}$ across all five transfer values. Every configuration uses five selection seeds and the same distinct-seed policy. Because a fixed low RR budget can overfit adaptive greedy selection, each CIM-RIS run starts with 100,000 nested conditioned samples and doubles the sample prefix. Its minimum terminal budget is the smaller of 1.5 million and the budget required to provide ten expected nonempty RR observations per coupon index and candidate, rounded upward to the next 50,000. After reaching this floor, we compare consecutive allocations on 5,000 fresh paired forward realizations and stop when their relative spread difference is at most 0.5%, or at a hard cap of 3 million samples. IC-RIS receives the same final RR count as CIM-RIS. Final results use 10,000 new forward realizations that are independent of both selection and stability validation. The comparison includes IC-RIS, 1Hop-Sort, Alpha-Sort, DegreeTopM, and PageRank. For each parameter cell, we compare CIM-RIS with the baseline having the highest mean spread in that cell; this is a stronger criterion than comparing against a fixed baseline.

Figure 8(a–b) reports the complete $k = 100$ grid. On NetFacebookEgo, CIM-RIS has higher mean spread than the strongest baseline in 19 of 20 cells, with a mean relative difference of 2.20% and a maximum of 4.75% at $(t, r) = (0.93, 0.80)$. The only negative cell is $(0.30, 0.20)$, where IC-RIS is 0.42% higher. On Netscience, CIM-RIS has the higher mean in 14 of 20 cells, with a mean difference of 0.75% and a range from -0.69% to 3.08% . Across both graphs, it exceeds DegreeTopM and PageRank in all 40 cells. Thus coupon-aware coverage is most beneficial when transfer is persistent, while simple adoption-aware rankings remain competitive in several low-transfer, low-redemption cells.

Figure 8(c–d) shows that these comparisons require substantially more than the earlier fixed low sample budgets. Across all 300 selection runs, the median final RR count is 300,000 and the mean is 534,500. The stopping rule is met in 295 runs; 11 reach the 3-million cap, of which five remain unstable. All five unresolved runs are retained in the reported aggregates. At $k = 100$, the largest mean terminal budget is 2.84 million in the NetFacebookEgo cell $(0.93, 0.20)$, consistent with the scarcity of root-adoption observations under high transfer and low redemption. These are empirical stability checks rather than the theorem-level (ϵ, δ) certificate.

The budget slice in Figure 8(g–h) further shows that the favorable region can depend on campaign scale. On NetFacebookEgo, CIM-RIS has the higher mean in one of five transfer settings at $k = 25$ and all five settings at both $k = 100$ and $k = 200$; the corresponding mean differences are -0.36% , 2.49% , and 2.65% . Netscience is less monotone: the corresponding win counts are two, four, and two, with mean differences of 0.82% , 1.08% , and 0.14% .

Finally, Figure 8(e–f) compares the duplicate-redemption fraction, defined as $(\bar{\rho} - \bar{\sigma})/\bar{\rho}$, with that of the strongest baseline. Positive values indicate that CIM-RIS converts a larger fraction of redemptions into distinct adopters. Across the 40 cells at $k = 100$, its Pearson correlation with relative spread difference is 0.43. Duplicate avoidance is therefore associated with the measured gain but does not fully explain it; whether the selected walks reach users who redeem also matters.

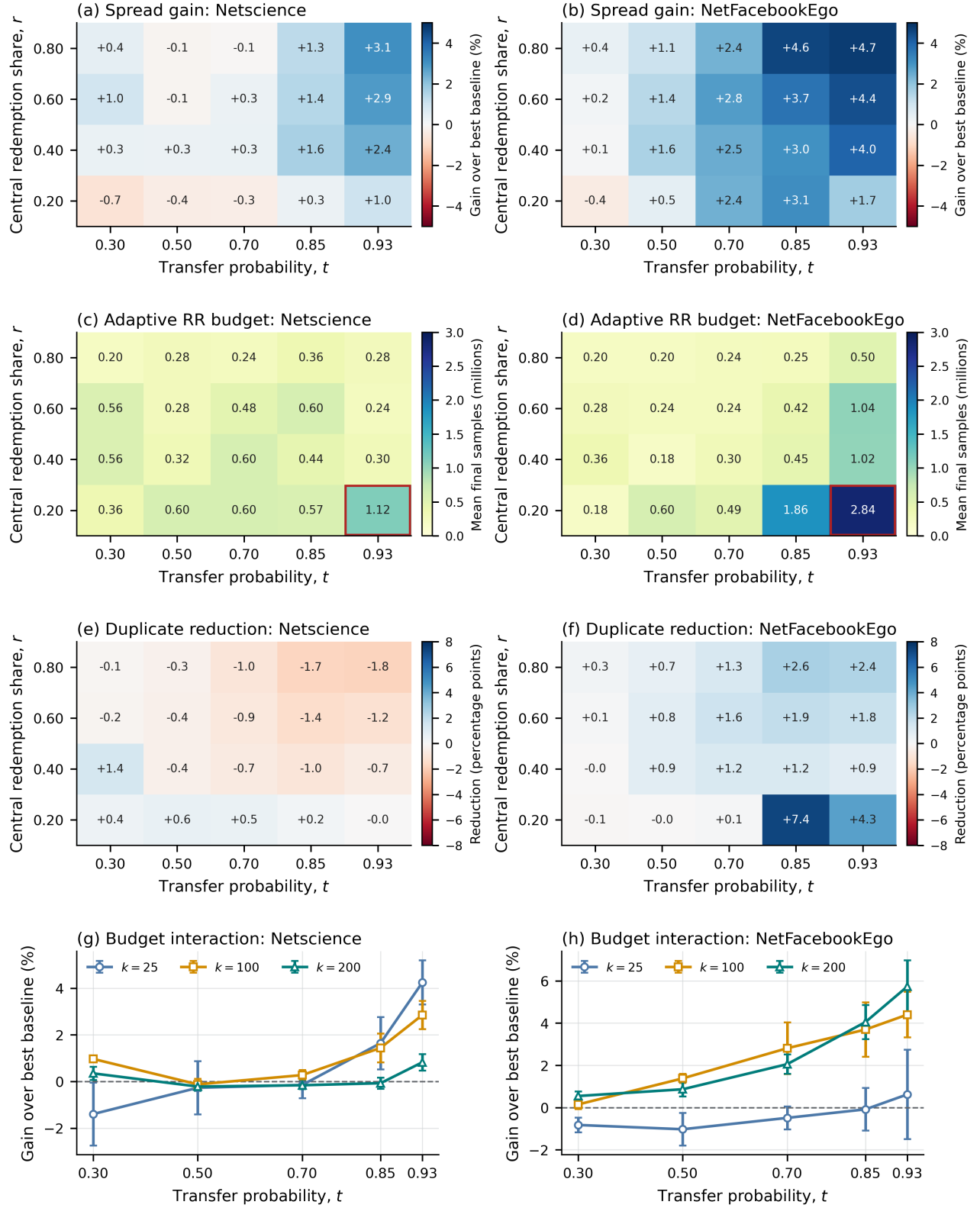


Figure 8: Complete parameter-regime study on two graph instances. (a–b) Relative adoption-spread difference between CIM-RIS and the strongest non-oracle baseline in each $k = 100$ cell. (c–d) Mean final RR sample count under adaptive stopping; red outlines mark cells containing at least one run that remained unstable at the hard cap. (e–f) Reduction in duplicate-redemption fraction relative to that same baseline; positive values favor CIM-RIS. (g–h) Relative spread difference for $r = 0.60$ as the budget changes. Heatmaps retain all prespecified cells, and line-chart error bars are 95% confidence intervals over five paired selection/evaluation runs.

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